

TrES-3b

R-1211

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_200502 · created 2026-07-31T07:49:00+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458971.7751 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.017
Transit depth (Rp/Rs)^2	3.85 +/- 0.66 [%]
Orbital Inclination (inc)	82.28 +/- 0.69
Airmass coefficient 1 (a1)	0.997 +/- 0.015
Airmass coefficient 2 (a2)	0.0029 +/- 0.0056
Scatter in the residuals of the lightcurve fit is	1.54 %
Best Comparison Star	#1 - [566, 331]
Optimal Aperture	4.0
Optimal Annulus	9.72
Transit Duration (day)	0.0618 +/- 0.0058

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.017 vs 0.16309 (deviation 1.84σ)
Duration fit vs published	0.0618 d vs 0.0567 d (deviation 0.84σ)
Mid-time O–C	-4.5 min
Depth SNR	10.9
Residual scatter	1.54 %

Light curve

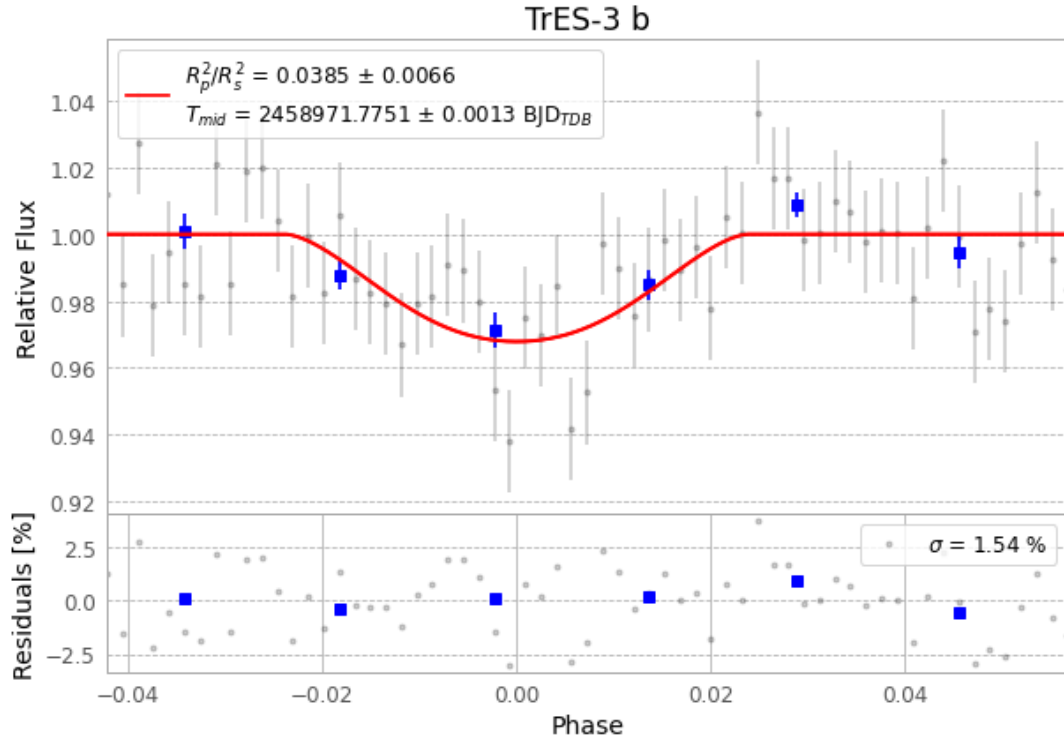


Figure 1 — FinalLightCurve_TrES-3 b_2020-05-01.png (SHA-256 f633787c0f555301...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [460, 174], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c6d8b1c6c2973e91...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

63 FITS frames (41 MB), TRES-3200502051212.FITS → TRES-3200502081812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-200502.log (f9f9b94c4c45...), FinalLightCurve_TrES-3 b_2020-05-01.png (f633787c0f55...), FinalLightCurve_TrES-3 b_2020-05-01.csv (10e8baa8f61c...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 07:49Z.